
FLEET FINANCIAL MANAGEMENT SYSTEM

Software Requirements Specification (SRS)

1. INTRODUCTION

1.1 Purpose

This document describes the requirements for a **Fleet Financial Management System** designed for a shipping company that owns and operates multiple ships. The system will automate financial tracking, reporting, and operational management.

1.2 Scope

The system will:

- Track all **income and expenses**
 - Manage **multiple ships**
 - Handle **employee salaries**
 - Track **loans and repayments**
 - Generate **financial reports (profit/loss)**
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2. SYSTEM OVERVIEW

The system is a **web-based application** that allows the company to:

- Record all transactions
 - Monitor ship performance
 - Track expenses and income
 - Generate reports per ship and overall
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3. OBJECTIVES

- Automate financial recording
 - Reduce manual errors
 - Improve reporting accuracy
 - Provide real-time financial insights
 - Support multiple ships independently
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4. USER ROLES

4.1 Admin

- Full system access
- Manage users
- Manage ships
- View all reports

4.2 Accountant

- Record income and expenses
- Manage loans
- Generate financial reports

4.3 Manager

- View reports only
 - Monitor ship performance
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5. SYSTEM MODULES

5.1 Fleet Management

Features:

- Add/Edit/Delete ships
- View all ships
- Assign employees to ships

Data:

- Ship Name
 - Registration Number
 - Status
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5.2 Income Management

Types of Income:

1. Sales (Cargo transport payments)
2. Delivery Orders (DO fees)

Features:

- Record income
- Filter by ship and date

Data:

- Amount
 - Type (Sales / DO)
 - Ship
 - Date
 - Description
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5.3 Expense Management

Categories:

- Marine Bills
- Salary
- Maintenance & Repair
- Fuel Cost
- Food & Grocery
- Port Cash
- ZMA Bills
- Loan Repayment
- Administration Cost
- Other Cost

Features:

- Record expenses

- Upload receipt
- Filter by category, ship, date

Data:

- Amount
 - Category
 - Ship
 - Date
 - Payment Method
 - Description
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5.4 Employee Management

Features:

- Add employees per ship
- Manage salaries
- Track salary payments

Data:

- Name
 - Role
 - Salary
 - Ship Assignment
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5.5 Loan Management

Features:

- Record loans
- Track repayments
- Update remaining balance

Data:

- Bank Name
- Total Loan Amount
- Monthly Payment
- Remaining Balance

5.6 Reporting System

Reports:

1. Profit & Loss

- Per ship
- Monthly
- Overall company

2. Expense Report

- Breakdown by category

3. Income Report

- Sales vs Delivery Orders

4. Loan Report

- Total paid
 - Remaining balance
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6. BUSINESS LOGIC

6.1 Profit Calculation

Profit = Total Income - Total Expenses

6.2 Ship-Based Calculation

Each ship must have:

- Its own income
 - Its own expenses
 - Independent profit calculation
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7. DATABASE REQUIREMENTS

Entities:

- Ships
- Employees
- Income
- Expenses
- Loans
- Loan Payments
- Sales
- Delivery Orders
- Users

Rules:

- Every record must include:
 - Ship ID
 - Date
 - Categories must be predefined (ENUM)
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8. AUTHENTICATION & SECURITY

- JWT Authentication
 - Role-based access control
 - Secure API endpoints
 - Input validation
 - Error handling
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9. API REQUIREMENTS

Ships

- Create, Read, Update, Delete

Income

- Add income
- Filter by ship/date

Expenses

- Add expense
- Filter by category/ship/date

Employees

- Manage employees
- Assign to ships

Loans

- Add loan
- Record repayments

Reports

- Profit/Loss
- Monthly reports
- Ship reports

10. FRONTEND REQUIREMENTS

Dashboard

- Total income
- Total expenses
- Profit/Loss
- Charts (monthly trends)

Pages:

- Ships Management
- Employees
- Income Entry
- Expense Entry
- Loan Tracking
- Reports

Features:

- Filters (date, ship, category)
- Responsive design
- Clean UI

11. NON-FUNCTIONAL REQUIREMENTS

Performance

- Fast response time
- Efficient database queries

Scalability

- Support multiple ships and users

Usability

- Simple and intuitive interface

Security

- Data protection
- Authentication required

12. FUTURE IMPROVEMENTS

- Mobile app version
- Notifications (loan due, high expenses)
- Integration with banking systems
- Advanced analytics

13. CONCLUSION

This system will provide:

- Full financial control
 - Accurate reporting
 - Better decision-making
 - Efficient ship management
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